

Retrospective & Contribution Report: PPOrphIt

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1 Team Contributions

We divided the team contributions into separate parts so that we can make progress much quicker and not be a blocker to each other as we move forward. Below are our contributions listed out.

Steve Gillet

- **Lead driver and architect:** Defined the research scope, hypothesis, and experimental design as part of his broader Master's Thesis work on morphological optimization. Primary driver of the project roadmap, Reinforcement Learning architecture, and PPO implementation.
- **Experimentation:** Designed and executed the core experimental loop, including baseline comparisons, reward shaping, and ablation studies.
- **Core RL Loop:** Implemented the core PPO training architecture using Stable Baselines3.
- **Reward Engineering:** Designed and tuned the multi-objective reward function (weighted sum of success, energy, and manipulability metrics).
- **Writing:** Lead author of the final paper.

Jay Vakil (Simulation, Evaluation & Optimization)

- **Simulation Architect:** Developed the multi-model generation pipeline in MuJoCo, enabling the environment to dynamically instantiate varying robot morphologies during training without crashing the physics engine.
- **Baseline Implementation:** Wrote a custom, scratch implementation of the REINFORCE algorithm (Policy Gradient) using PyTorch to serve as a sanity check and additional baseline for the primary PPO agent.
- **Ablation studies:** Assisted with running evaluation scripts and debugging the integration between the PPO agent and the physics environment.
- **Visualization:** Produced the morphology and evaluation plots used in the final report and presentation.
- **Support:** Provided supporting implementation for the RL loop and assisted with evaluation scripts.

2 Post-Mortem

2.1 What Worked

- **Dynamic XML Generation:** A major technical risk was whether MuJoCo could handle reloading and spawning different robot descriptions on the fly. Our wrapper successfully handled this by re-initializing the physics engine only when morphology parameters changed, maintaining training stability.
- **The “Reachability” Discovery:** The evaluation revealed that for our specific linear tasks, the Panda’s fixed geometry made it physically impossible to reach the goal in certain workspace configurations. This unintentionally but effectively validated our core thesis: fixed robots are often ill-suited for arbitrary tasks.

2.2 What Failed / Technical Hurdles

- **Reward Exploitation & Reality Gap:** A significant issue arose from the agent optimizing strictly for the provided reward function rather than physical plausibility. As seen in the generated “Container Task” morphologies, the agent favored long slide joints to maximize the Manipulability Index. While this boosted the score, these directional slide joints are often impractical for general-purpose manufacturing. Furthermore, the agent frequently converged on designs with small base joints supporting massive, fully extended arms that moved in large overhead arches. In reality, the torque required to hold these extensions against gravity would be prohibitive, but our initial reward function did not sufficiently penalize mechanical effort.
- **Energy Constraint Implementation:** To address the issue above, we designed a “Wall Task” environment (mounting the arm sideways to increase gravitational complexity) and attempted to incorporate an energy penalty using MuJoCo’s inverse dynamics ($\text{torque} \cdot \text{velocity}$). However, accurately tuning the weights for this energy term without destabilizing the PPO training proved difficult, and we could not fully validate this corrected reward function within the project timeline.
- **Training Duration:** PPO training took significantly longer than anticipated. Convergence on the morphology search space required millions of timesteps, making the feedback loop for testing new reward shapes (like the energy penalty mentioned above) extremely slow.
- **Dreamer Integration:** We originally proposed comparing PPO against the Dreamer (World Model) architecture. Integrating Dreamer’s latent state requirements with our dynamic observation space proved too complex for the timeline, forcing us to pivot solely to PPO.

3 Hypothetical Next Steps (3-Month Plan)

With three additional months, we would expand the project in the following directions:

1. **Implement DreamerV3:** We would revisit the model-based RL approach. We hypothesize that a world model would learn the “morphology-to-performance” mapping significantly faster than PPO, drastically reducing simulation steps.
2. **Hardware-In-The-Loop:** We would use rapid prototyping (3D printing) to physically instantiate the PPO-optimized links and assemble them with standard Dynamixel servos. This would validate if the “Manipulability Index” gains in simulation translate to real-world dexterity.

3. **Complex Manipulation Tasks:** We would expand beyond pick-and-place to “Peg-in-Hole” assembly tasks. This would likely force the agent to evolve more precise, stiff morphologies rather than the long reaching arms observed in the current study.